

Executive HR Employee Attrition Report

Employee Turnover Prediction & People Analytics

Dataset: 1470 employees

Overall attrition rate: 16.12%

Selected predictive model: Logistic Regression

Executive Summary

This analysis uses employee demographic, job, satisfaction, income and work-related information to identify patterns associated with employee attrition. A machine learning classification model was developed to estimate the probability that an employee may leave the organization.

The model should be used as a decision-support tool rather than as a final HR decision system. Predictive relationships do not necessarily prove that a feature directly causes employee turnover.

Model Performance

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.76	0.36	0.66	0.46
Random Forest	0.84	0.48	0.32	0.38

Top 3 Predictive Drivers

- OverTime
- NumCompaniesWorked
- BusinessTravel

Department Breakdown

Department	Employees	Attrition	Rate
Sales	446	92	20.63%
Human Resources	63	12	19.05%
Research & Development	961	133	13.84%

Data-Backed Retention Recommendations

- Use the model risk score to identify employees who may benefit from an early retention conversation.
- Investigate workload and overtime patterns and consider workload balancing where appropriate.
- Strengthen manager-employee communication and career-development discussions for employees showing elevated risk.
- Monitor satisfaction and engagement indicators regularly, particularly in departments or roles with higher predicted attrition.

- Use compensation and career-growth reviews as part of retention discussions where income and progression patterns are associated with elevated risk.

Risk Watchlist

The model identified 236 employees with a predicted high attrition risk (risk score $\geq 70\%$). The detailed watchlist is available in the project output CSV file.

Important Limitations

Risk scores are predictions based on historical data and should not be treated as certainty. HR teams should review the underlying context before taking action. The model should support human judgment rather than replace it.